INTERNATIONAL UNIVERSITY OF EAST AFRICA
'Learning to Succeed'

FACULTY OF SCIENCE AND TECHNOLOGY

LOGIC BUILDING PROGRAMMING & VERSION CONTROL REPORT

PROGRAMME: BSSE/BSCS/BIT

YEAR/SEM: YEAR 2 SEM 1

COURSE CODE: BIT2115

PROBLEM NAME: JAVA ACCESS MODIFIERS

STUDENT DETAILS:

NAME OF STUDENT	**Mulumeoderhwa Bahati Dieudonne**
REG NUMBER	**25/1512/BSCS-S**

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Section 1

Logic Building & Programming

Lines of code & explanations

Line / Part	Explanation
package patient;	Organizes the Patient class inside the patient package.
private String medicalRecord;	Protects medical information from direct external access.
private String financialRecord;	Protects sensitive financial information.
public String patientName;	Allows authorized classes to access the patient's basic name information.
public String appointmentDate;	Allows access to the patient's appointment date.
public Patient(...)	Creates and initializes a patient object.
public void updateMedicalRecord(...)	Allows a doctor to update the medical record through a controlled method.
public String getMedicalRecord()	Provides controlled access to the medical record.
private String getFinancialRecord()	Keeps the financial record inaccessible directly from other packages.
package staff;	Organizes staff-related classes into the staff package.
Doctor	Represents a doctor who needs access to medical information.
Receptionist	Represents a receptionist who only views basic patient information.
package administration;	Organizes administration-related classes.
HospitalAdmin	Represents hospital administration personnel.
import patient.Patient;	Allows classes in other packages to use the Patient class.
Patient patient = new Patient(...)	Creates a patient object containing the patient's information.

doctor.viewAndUpdatePatient(patient);	Gives the doctor access to the operations provided for medical information.
receptionist.viewBasicDetails(patient);	Allows the receptionist to view only basic patient details.
admin.managePatient(patient);	Calls the administration operation for the patient.

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Detailed Screenshots and caption descriptions

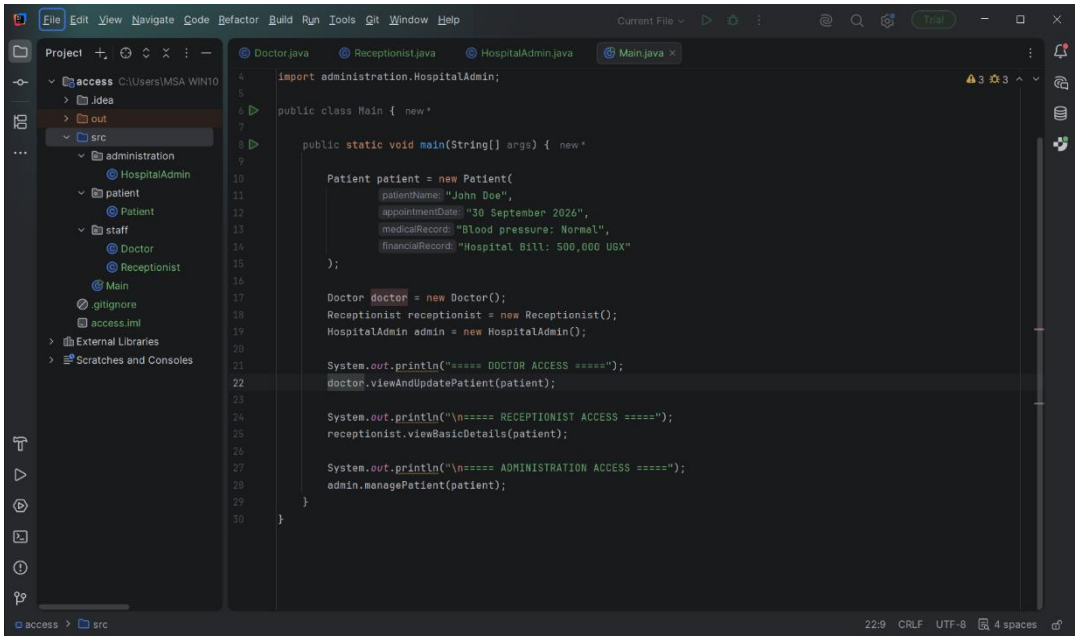


Figure 1:Class Main showing project structure

Output and caption descriptions

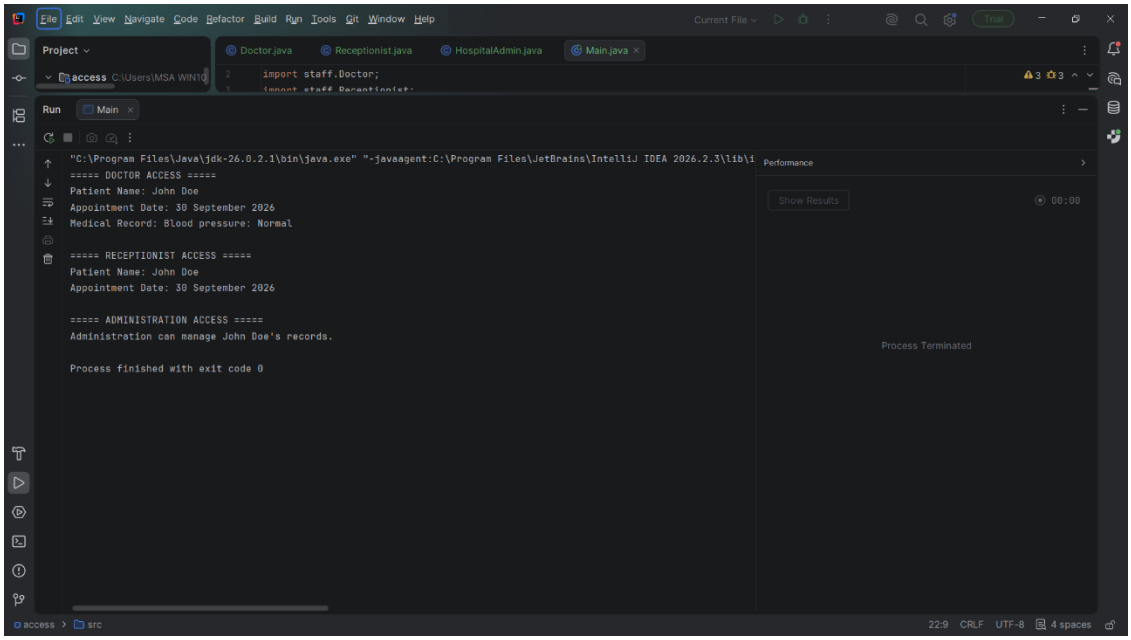


Figure 2: Output after running the program

Output Explanation

When the program runs, the doctor can view and update the patient's medical information, while the receptionist can only view the patient's name and appointment date. Sensitive financial information is kept private and cannot be accessed directly from outside the Patient class. This demonstrates how packages and access modifiers can control visibility and protect sensitive data.

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Section 2

Version Control

Repository Overview

Repository Name	**OOP**
Repository URL	**https://github.com/dieudonne-ai/OOP.git**
Branches	**2**

Commit #1 On Branch #1 (Main)

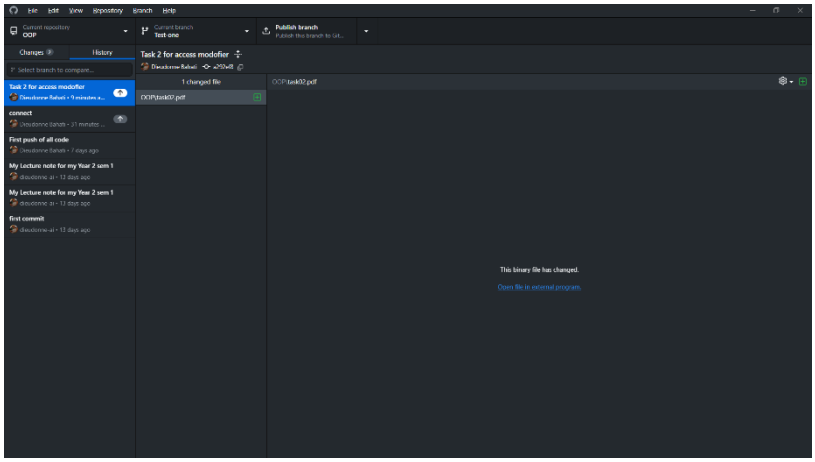


Figure 3: Commit 1 on Branch 1

Commit Summary

Commit Name	**First commit**
Commit ID	**a292ef8405d2f4f2a2b66f8d07d68d64b579399c**
Commit Description	
Branch Affected	**Test one**

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Commit #2 On Branch #2

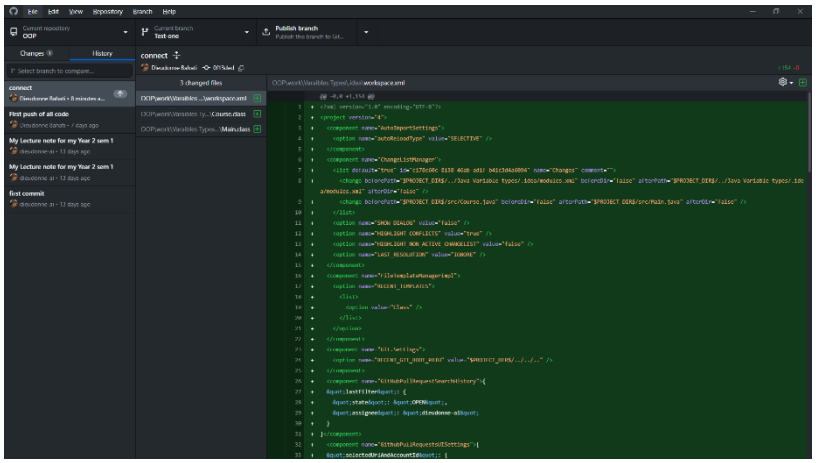


Figure 4: Commit 1 on Branch 1

Commit Summary

Commit Name	**New connection**
Commit ID	**9a278a2806e94f25f2f041ae7b24ef5c081c1536**
Commit Description	
Branch Affected	**Test-one**

Section 3

What Have You Learned From The Task?

Skill Set	**Description**
Packages	Learned how to organize Java classes into different packages.
Access Modifiers	Learned how public and private control access to classes, variables, and methods.
Data Protection	Learned how private variables can protect sensitive patient information.
Encapsulation	Learned how methods can provide controlled access to private data.
Class Organization	Learned how different hospital roles can be represented using separate classes.

Access Control	Learned how different users can be given different levels of access to information.